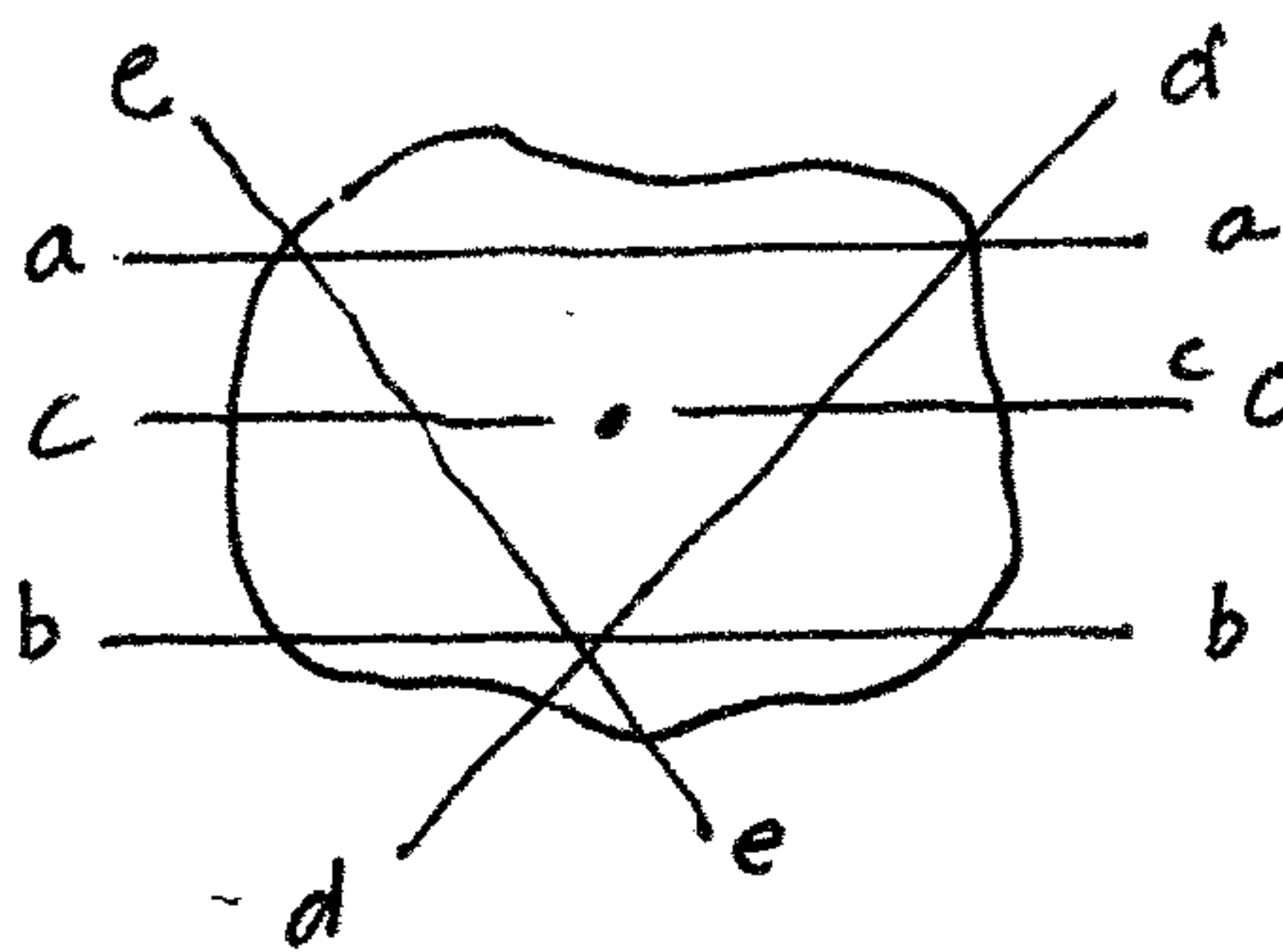


Parallel Axis Theorem (PAT): Derivation

(from Hibbeler Ch. 10.2)



At left is a general area, A.

It has a different (Area) moment of inertia

about every axis:

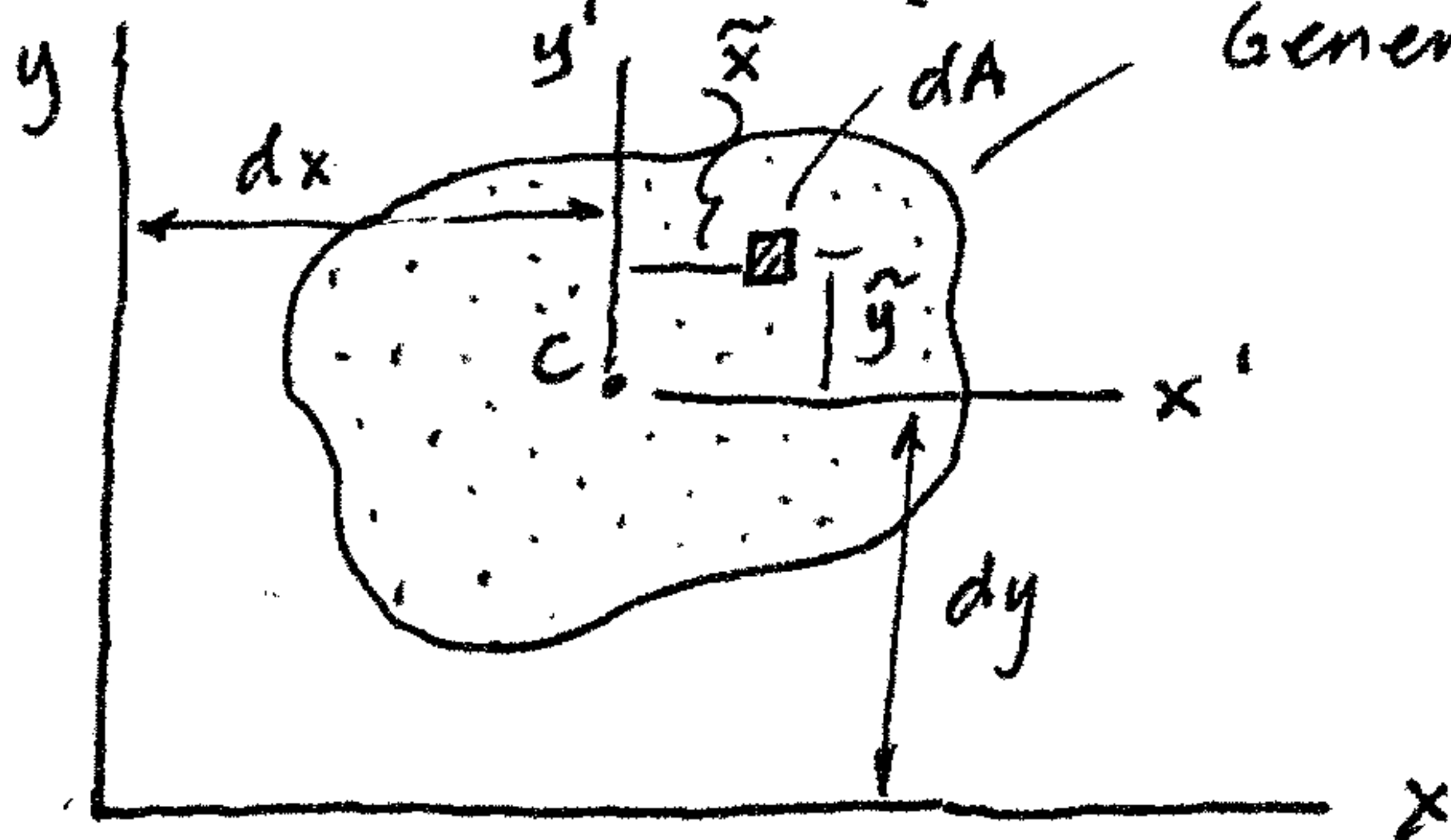
About axes parallel to others,

and about rotated axes.

$\left\{ \begin{array}{l} a-a \\ b-b \\ c-c \\ d-d \\ e-e \end{array} \right.$

In order to determine the MOI of an area about an axis that is parallel to the centroidal axis, we use the Parallel Axis Thm. (I call it "PAT")

To derive this:



General Area

x' and y' are centroidal axes

Parallel to x and y.

$$I_x = \int_A (dy + \tilde{y})^2 dA$$

This is $\bar{I}_{x'}$

$$= \int_A \tilde{y}^2 dA + 2dy \int_A \tilde{y} dA + dy^2 \int_A dA$$

Since $\tilde{y}$ are measured from centroidal axis

Area

Thw, $I_x = \bar{I}_{x'} + A dy^2$

- Book

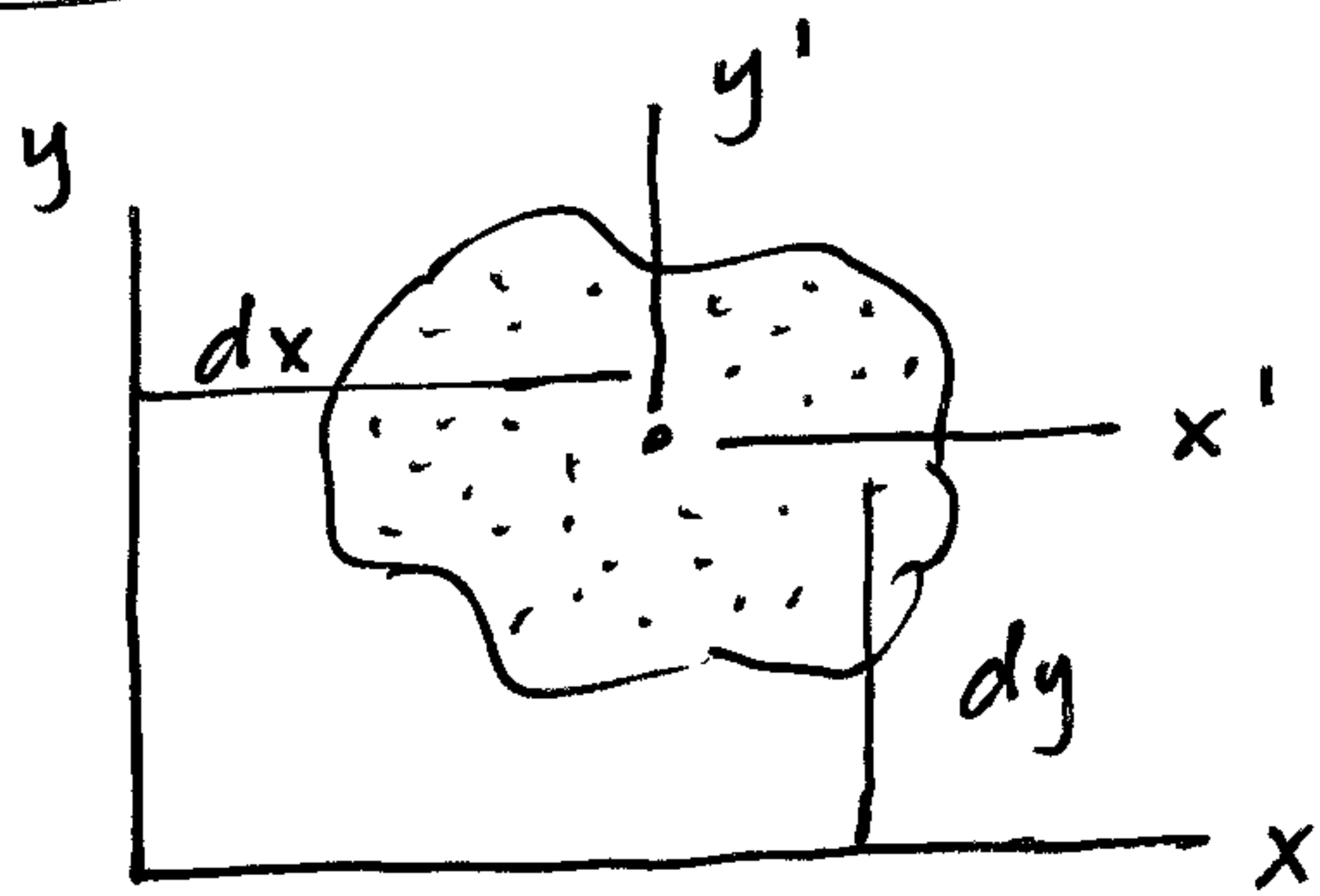
I write it this way.

or

$$I_x = \bar{I} + A dy^2$$

$\bar{I}$ implies centroid, so the x' subscript is redundant.

Derivation of Parallel Axis Theorem (PAT) (contd)



In deriving the PAT, we get:

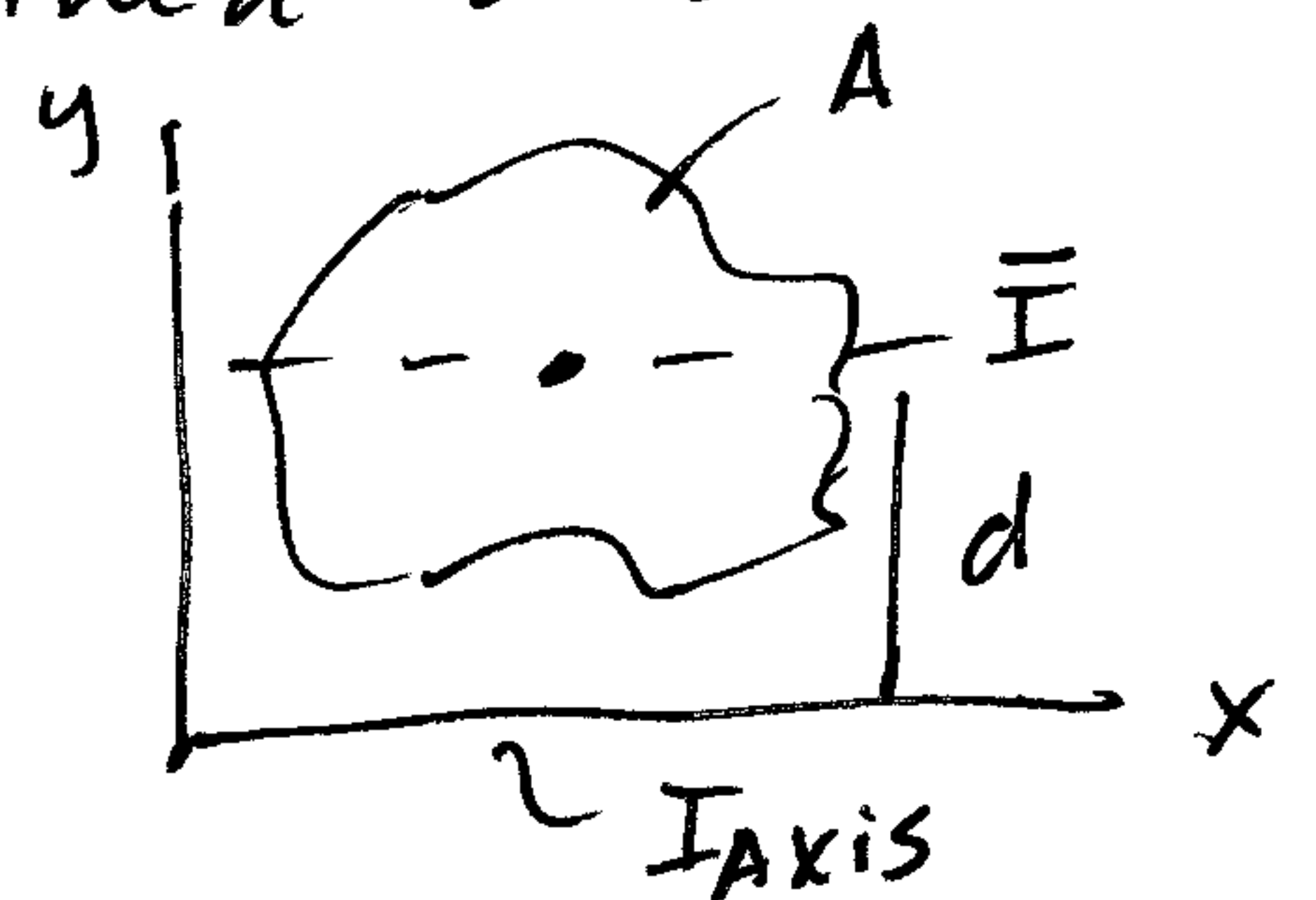
$$I_x = \bar{I}_{x'} + A dy^2$$

and

$$I_y = \bar{I}_{y'} + A dx^2$$

I prefer to generalize the formula as:

$$I_{Axis} = \bar{I} + A d^2$$



Where it is understood that ...

I_{Axis} -- I about an axis

$\bar{I}$ -- Centroidal I

A -- Area

d -- $\perp$ distance between the centroidal axis (for $\bar{I}$) and the axis of interest (I_{Axis})

For Composite Shape:

$$I_{Axis} = \sum (\bar{I} + A d^2)$$